483

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Development of High-Yielding Flood-Resilient Rice Varieties for Rainfed Lowland Ecosystem

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Rice is the primary food source for millions of small land-holding farmers in coastal regions. However, rice production in these areas, particularly in Asia and Africa, faces significant challenges due to frequent submergence and stagnant flooding. These factors severely limit rice yields and threaten the food and nutrition security of vulnerable smallholding farmers in flood-prone regions. To address this issue, an effective breeding strategy is crucial for developing flood-resilient rice varieties that are high-yielding and nutritious, capable of adapting to various conditions, including submergence and stagnant flooding. Past breeding efforts to create and disseminate submergence tolerance lines combined with stagnant flooding have been slow and ineffective. To overcome these limitations, we have recently developed a novel breeding approach at IRRI (International Rice Research Institute) that integrates the screening procedure for submergence and stagnant flooding into a unified background. Our innovative breeding approach enables us to improve both submergence and flooding tolerance simultaneously, resulting in the development of climate-resilient rice varieties with enhanced resilience to flooding. In this study, we will discuss the details of our screening procedure and outline the steps involved in integrating submergence and flooding tolerance traits into a single background. Our research aims to foster the development and dissemination of climate-resilient flooding rice varieties, ensuring food security and improving the livelihoods of smallholding farmers in flood-prone ecologies.